

Fifth Disease (Erythema Infectiosum)

Fifth disease, the most common clinical manifestation of parvovirus B19 infection, typically begins with nonspecific symptoms such as headache, coryza, and low-grade fever, occurring approximately 2 days before the onset of the rash. During the rash phase, patients may experience additional symptoms including headache, pharyngitis, fever, malaise, myalgias, coryza, diarrhea, nausea, cough, and conjunctivitis.

Approximately 10% of children with erythema infectiosum develop arthralgias or arthritis. Large joints are more commonly affected than small joints. In rare cases, children may present with chronic joint complaints that resemble juvenile rheumatoid arthritis.

Clinical Presentation of the Rash

The characteristic rash begins with bright red, confluent, erythematous, and edematous plaques on the malar eminences, producing the classic "slapped cheek" appearance. As the facial rash fades over 1 to 4 days, a secondary rash appears on the trunk, neck, and extensor surfaces of the extremities. This rash consists of pink to erythematous macules or papules with central clearing, giving it a lacy or reticulated appearance.

The rash may present in various morphologies, including morbilliform, confluent, circinate, or annular patterns, and involvement of the palms and soles has been reported. The eruption typically lasts 5 to 9 days but may recur intermittently for weeks or even months, triggered by factors such as sunlight, exercise, temperature changes, bathing, or emotional stress. Pruritus can be a prominent feature in some pediatric cases.

Other Manifestations

Parvovirus B19 has been occasionally associated with vascular purpura, including Henoch-Schönlein purpura. An enanthem may also occur,

characterized by erythema of the tongue and pharynx, along with red macules on the buccal mucosa and palate.

Parvovirus B19 in Adults

In adults, acute arthropathy is the predominant manifestation of parvovirus B19 infection. It occurs more frequently in women and primarily affects the small joints of the hands and knees, although other joints such as those in the spine and costochondral joints may also be involved. The arthritis is typically symmetric, of sudden onset, and self-limited, though it can persist or recur for months. It may clinically mimic Lyme arthritis or rheumatoid arthritis.

Constitutional symptoms tend to be more severe in adults than in children. The typical presentation includes fever, adenopathy, and mild arthritis, often without a rash. Women are more likely than men to develop joint symptoms and rash, whereas men frequently present with a flu-like illness alone. Some adults experience prolonged fatigue, malaise, and depression following infection.

Asymptomatic infections occur in both adults and children; in one outbreak, 26% of infected adults were asymptomatic. Parvovirus B19 can also cause paresthesias, such as numbness and tingling of the fingers, with or without other signs of fifth disease. Pruritus, sometimes severe, may occur with or without a rash. Notably, the presence of pruritus in a patient with acute-onset arthritis should prompt consideration of parvovirus B19 infection as a potential cause.